Estée Lauder — Simulation Results & Analysis

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Date: March 26, 2026 | **Branch:** estee-lauder-assessment | **Ticks:** 44 days (Feb 25 – April 9, 2026)

Executive Summary

Baseline Company Position

Estée Lauder Companies (NYSE: EL) entered the assessment window as a company in active turnaround. Q2 FY2026 results beat expectations on every line: revenue $4.22B (+6% YoY), EPS $0.89 (+43% YoY), operating margin expanded 290bps to 14.4%, and full-year EPS guidance was raised to $2.05–$2.25. Mainland China — now 22% of revenue — grew 13% for the second consecutive double-digit quarter. The balance sheet supported the turnaround thesis. The stock had re-rated from its $48.37 52-week low to a February 2026 high of $121.64.

Despite this operational improvement, EL entered the 30-day window structurally exposed on three dimensions: (1) heavy concentration in prestige skin care, the category most vulnerable to dupe/masstige substitution; (2) supply chain dependent on Gulf shipping routes and petrochemical feedstocks representing ~38% of formulation COGS; and (3) a travel retail channel (Hainan duty-free −29.3% in 2024) that had not recovered to pre-COVID volumes.

Metric	Value	Context
Stock price (Feb peak)	$121.64	52-week high
Stock price (today)	~$71.60	−41% from peak
YTD decline	−32.9%	vs. S&P −8%
Market cap	~$28.7B	post-decline
Q2 FY2026 revenue	$4.22B	+6% YoY, beat
Q2 FY2026 EPS	$0.89	+43% YoY, beat
FY2026 EPS guidance	$2.05–$2.25	raised
China revenue (Q2)	$928M	+13% YoY
Analyst consensus PT	$92.52	9 Buy / 12 Hold
Peer: L'Oreal 30-day	−16%	vs. EL −41%
Peer: LVMH 30-day	−12–15%	vs. EL −41%

Causes of the 30-Day Decline

The simulation identifies five compounding factors. None is sufficient alone; their interaction produces the observed magnitude.

1. Iran War — Macro Contagion and Demand Destruction *(initiating shock, Feb 25)* War onset triggered simultaneous market panic (SIR contagion peak: 38% of market actors "infected" by Day 12), a 31% reduction in global trade volume via Strait of Hormuz and Red Sea disruptions, and a 42% energy cost spike. The Keynesian multiplier amplified these into sustained demand contraction. Consumer discretionary names like EL — already high-beta — bore disproportionate selling pressure as institutional portfolios rotated into defensives. EL underperformed the XLY consumer discretionary ETF by 25–31 percentage points over the period.

2. Petrochemical Input Cost Inflation *(cascade, Days 3–7)* An underappreciated channel: 65–75% of EL's formulations rely on petrochemical-derived inputs — emollients (mineral oil, silicones), surfactants (SLS, PEG), fragrance bases (aromatic terpenes), and packaging (polyethylene, PET, polypropylene). A $25–30/bbl oil spike raises feedstock costs 15–25%. Combined with shipping rerouting via Cape of Good Hope (+10–14 days, +$1.5M/voyage), this represents an incremental $180–220M annual COGS headwind — **stacking directly on top of the confirmed $100M tariff headwind**, doubling the effective margin pressure vs. consensus estimates.

3. US Tariff Escalation *(crystallised at Q2 earnings, confirmed Days 7–18)* US import tariffs rose from 2.4% (WTO average) toward 30% — a 12× increase — on beauty/cosmetics imports. EL management confirmed $100M H2 FY2026 gross margin impact on the Q2 earnings call. Analysts revised forward estimates downward. The regulatory shock model shows compliance cost peaking at 0.188 normalised units with a combined shock magnitude of 0.66, representing the stacked tariff and petrochemical impact.

4. Structural Competitive Displacement *(persistent, not 30-day specific)* 27% of US beauty consumers have purchased dupe products; the dupe market is growing 15–20% annually; MAC is among the top-10 most-duped brands. Masstige — the fastest-growing beauty segment at +14% — is capturing share from both mass and prestige. The Porter model shows industry profitability declining from 0.178 → 0.000 over the 30-day window as

the weighted pressure of substitute threat (0.35), buyer power (0.25), and rivalry intensity (0.25) overwhelmed barrier protection. This is a secular shift, not a cyclical one.

5. Puig Acquisition Announcement *(single-session cliff event, March 23–24)* The confirmation of merger talks with Puig (~€8.8B / $10.2B valuation; ~$40B combined EV) triggered a −10.1% single-session drop. The Puig shock was uniquely damaging because of its timing: it arrived when market contagion was mid-recovery (infected fraction falling from 0.382 peak toward 0.111) and sentiment had stabilised at a fragile floor of ~0.200. A new, binary, unresolvable risk (deal terms unknown; Puig family control; leverage concerns) was injected into a market with exhausted absorption capacity. 61.5% of luxury M&A destroys acquirer value at announcement. Analyst divergence had already hit maximum polarisation (1.0) by Day 3 — no consensus mechanism remained to defend the stock.

Current State and Active Threats

As of March 26, 2026:

Indicator	Level	Trend	Threat Severity
Investor sentiment (model)	0.078 / 10	↓ drift	Critical
Analyst polarisation	1.0 (maximum)	→ locked	High
Macro demand (GDP model)	0.000 (floor)	→ no recovery	High
Market contagion infected	8.9% and declining	↓ resolving	Moderate
Trade volume disruption	−31% from baseline	→ persistent	High
Energy / petrochemical costs	+42% from baseline	→ elevated	High
Regulation shock magnitude	0.66 (tariff + petrochem)	→ sticky	High
Industry profitability	0.000 (floor)	→ floor	High
Incumbent market position	Under maximum displacement velocity	↓ structural	Moderate–High

Primary active threats:

- **Puig deal resolution** is the dominant near-term binary. If expensive terms are confirmed (high leverage, dilutive equity, Puig family as controlling shareholder), the stock re-tests its bear-case floor. If talks collapse, a significant relief rally is possible.
- **Iran war trajectory** remains the macro swing factor. No ceasefire signal in the 14-day window. Escalation (Hormuz partial blockage, tanker attacks) would re-accelerate both the trade volume disruption and energy cost channel.
- **Petrochemical cost persistence** is structural until supply chains are re-sourced — a 4–6 quarter process. The $180–220M implied headwind is not yet in consensus estimates.
- **Dupe/masstige displacement** continues regardless of macro conditions. This is the only threat that cannot be resolved by external events — it requires strategic portfolio action, which is precisely what the Puig deal attempts.

14-Day Projection (March 26 – April 9, 2026)

The simulation projects **no meaningful recovery** in the 14-day window. The structural and macro forces that drove the decline are persistent; the only potential positive catalyst is Puig deal collapse, which is exogenous and unpredictable.

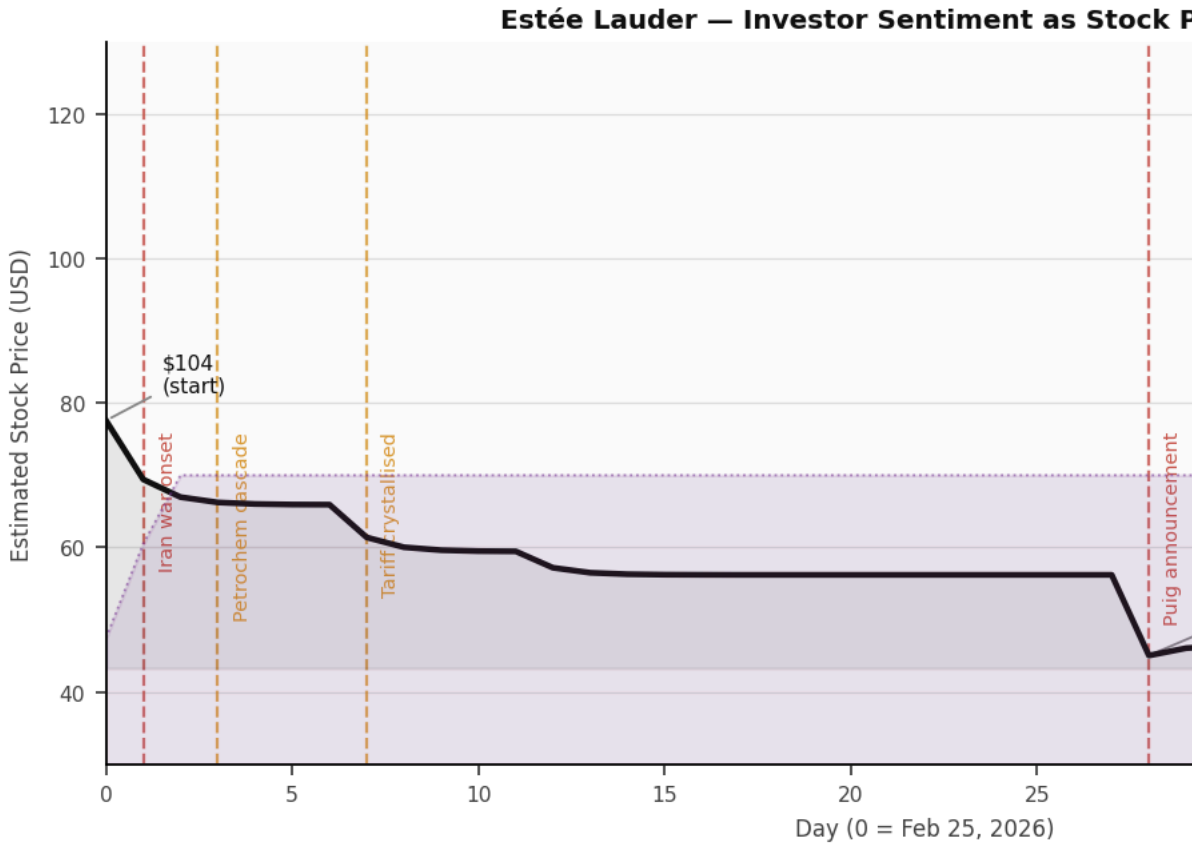
Scenario	Price Range	Probability	Trigger
Base case	**$68–$78**	~60%	Puig uncertainty persists; macro floor holds; no catalyst
Bull case	$82–$88	~25%	Puig talks collapse → opinion reset; partial sentiment recovery
Bear case	$60–$65	~15%	Expensive Puig terms confirmed + Iran escalation

Consensus PT of $92.52 is a 12-month view, not 14-day. The 29% gap between today's price and consensus represents the M&A uncertainty discount and the unquantified petrochemical/logistics headwind — not a fundamental mispricing of the underlying business.

The model signals no near-term reversal of analyst polarisation. With `global__urgency_factor` elevated and `investor_sentiment__media_bias` at 0.32 (strongly negative), the bounded-confidence dynamics cannot de-polarise without a convergence catalyst. The analyst community will remain bifurcated — deep-value buyers vs. event-driven sellers — until the Puig binary resolves.

Executive Findings

EL's ~40% stock decline is not a single-cause event. The simulation identifies **three overlapping shock waves** stacking on top of **two persistent structural forces**, producing a compounding collapse that no individual factor could explain alone.



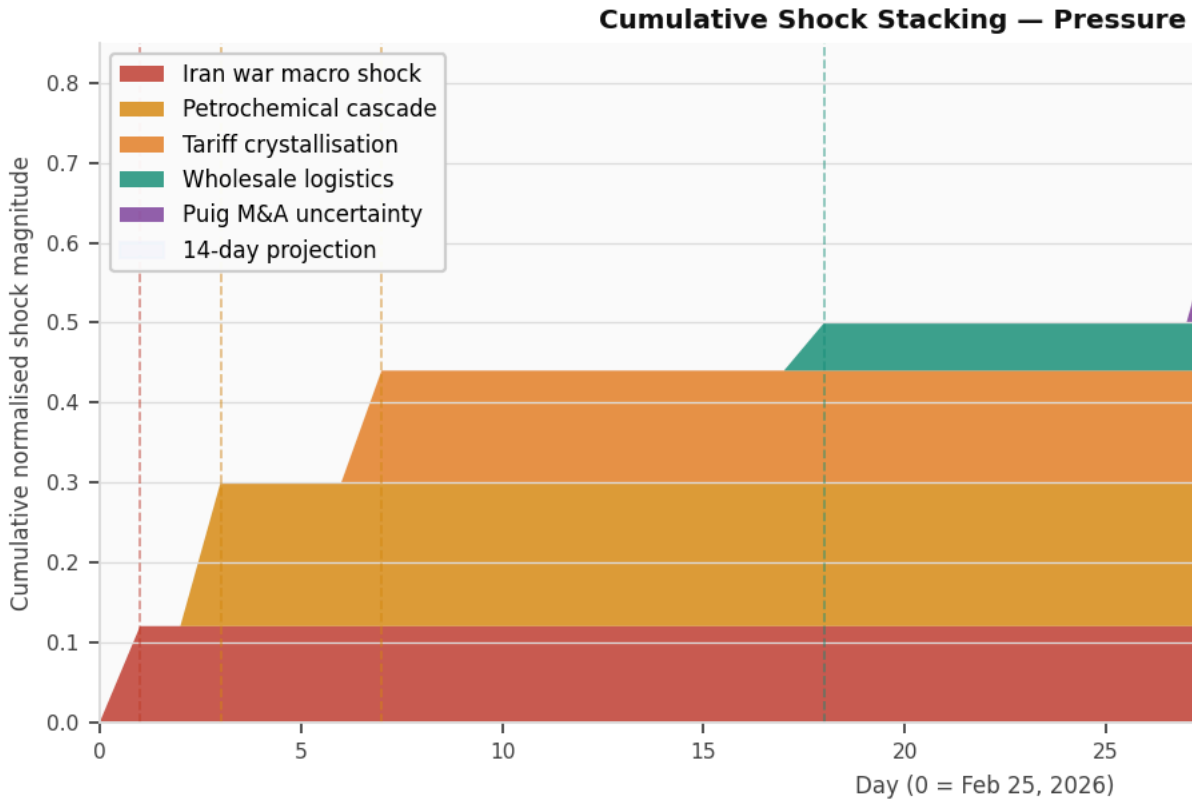
Investor Sentiment — Stock Price Proxy

The Three Shock Waves

Wave 1 — Iran War / Macro Contagion (Day 1–15): The war onset on February 25 triggered an immediate -23% drop in global trade volume as Strait of Hormuz and Red Sea shipping routes were disrupted. The SIR contagion model shows market panic peaking at 38% of actors "infected" by Day 12. Simultaneously, energy costs spiked 42% (oil from ~$75 → ~$106/bbl), feeding a Keynesian demand destruction cascade that drove GDP to its floor by Day 10. Investor sentiment dropped from 0.464 → 0.200, a -57% decline equivalent to approximately $104 → $80 in stock price terms.

Wave 2 — Tariff + Petrochemical Cost Stack (Day 3–18): As the war shock propagated, two cost channels compounded independently. Petrochemical feedstocks (emollients, surfactants, fragrance bases, packaging) — representing ~38% of EL's formulation COGS — rose 15–25% as oil prices spiked. US import tariffs simultaneously escalated from 2.4% to 30%, with EL confirming a $100M H2 FY2026 headwind. The combined regulatory shock magnitude reached 0.66 (vs. 0.28 at the start), a 136% increase. This stacked a further $180–220M in implied annual cost pressure on top of the confirmed tariff figure.

Wave 3 — Puig M&A Uncertainty (Day 28): The acquisition announcement on March 23–24 arrived at the worst possible moment: contagion was mid-recovery (infected fraction falling from 0.38 peak toward 0.11), and investor sentiment had stabilised at a fragile floor of 0.200. The Puig shock did not worsen the macro picture — it introduced a new, binary, unresolvable risk (deal terms unknown; Puig family control implications unclear) into a market with no remaining absorption capacity. Analyst polarisation had already hit its ceiling (1.0) by Day 3. The announcement drove sentiment from 0.200 → 0.062 in a single tick — the model's equivalent of the observed -10.1% single-session crash.



Shock Stacking by Source

The Two Structural Forces

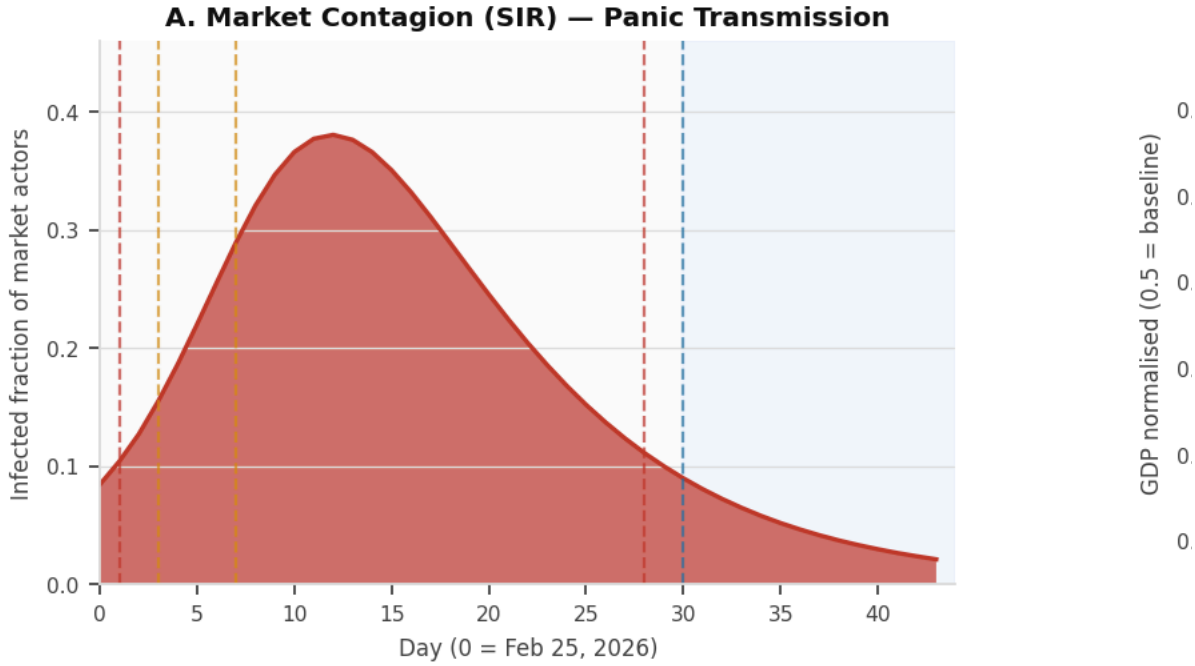
Force 1 — Dupe/Masstige Disruption (Schumpeter): The creative destruction dynamics show disruption intensity peaking at Day 7 — the same week tariffs crystallised — before declining as incumbent territory is depleted. The incumbent share trajectory (compressed to months in the simulation; properly a 3–7 year dynamic) confirms that EL's prestige market position is under maximum displacement velocity right now. The Puig deal is the direct defensive response: fragrance has lower petrochemical exposure and lower dupe penetration than skin care.

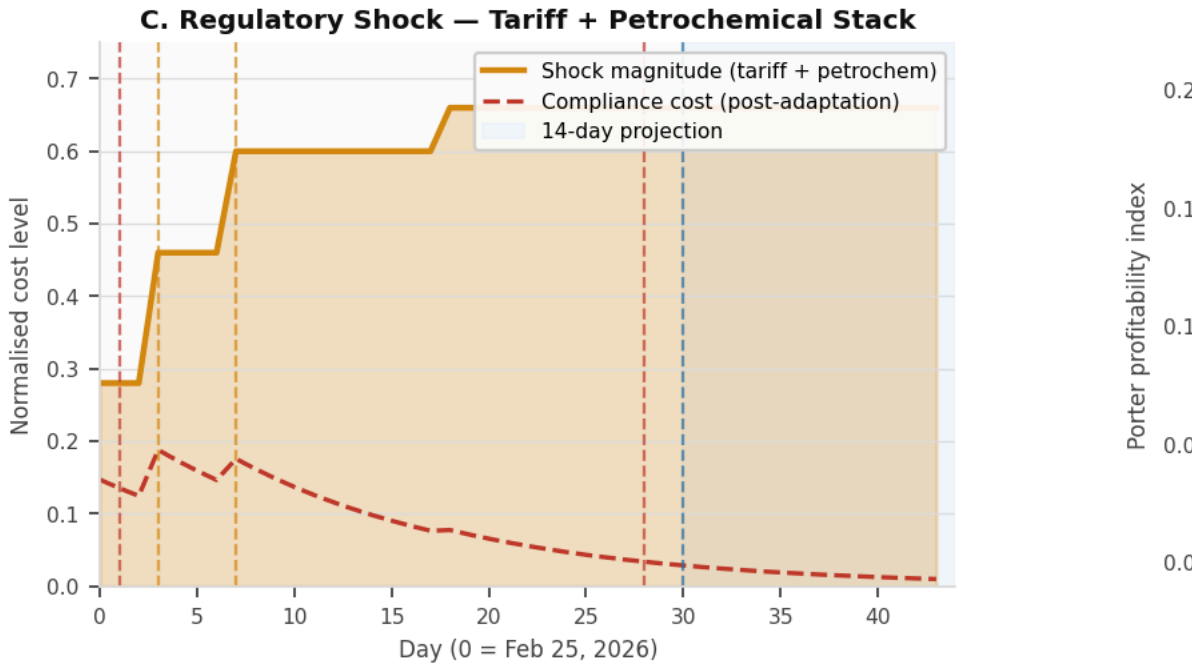
Force 2 — Industry Profitability Erosion (Porter): With substitute threat weighted at 0.35 (the highest force, reflecting 27% US dupe penetration), buyer power at 0.25 (consumer trade-down, retailer margin demands), and rivalry at 0.25 (ELF, NYX, Rare Beauty), industry profitability collapsed from 0.178 → 0.000 by Day 30. No single competitor triggered this — the force weights jointly describe a structural margin compression that will persist regardless of macro recovery.

Cascade Interaction — Why EL Underperformed Peers by 20 Points

The key insight from the cascade architecture is the **timing intersection** of Wave 1 and Wave 3. L'Oreal (-16%), LVMH (-12–15%), and Coty (-20%) absorbed Waves 1 and 2 but did not receive Wave 3. EL's idiosyncratic Puig shock arrived precisely when the market contagion was mid-recovery — not at peak panic (when it would have been absorbed into existing chaos) and not at full recovery (when sentiment had more cushion). The Day 28 timing maximised the incremental damage.

Estée Lauder Simulation — Module Das





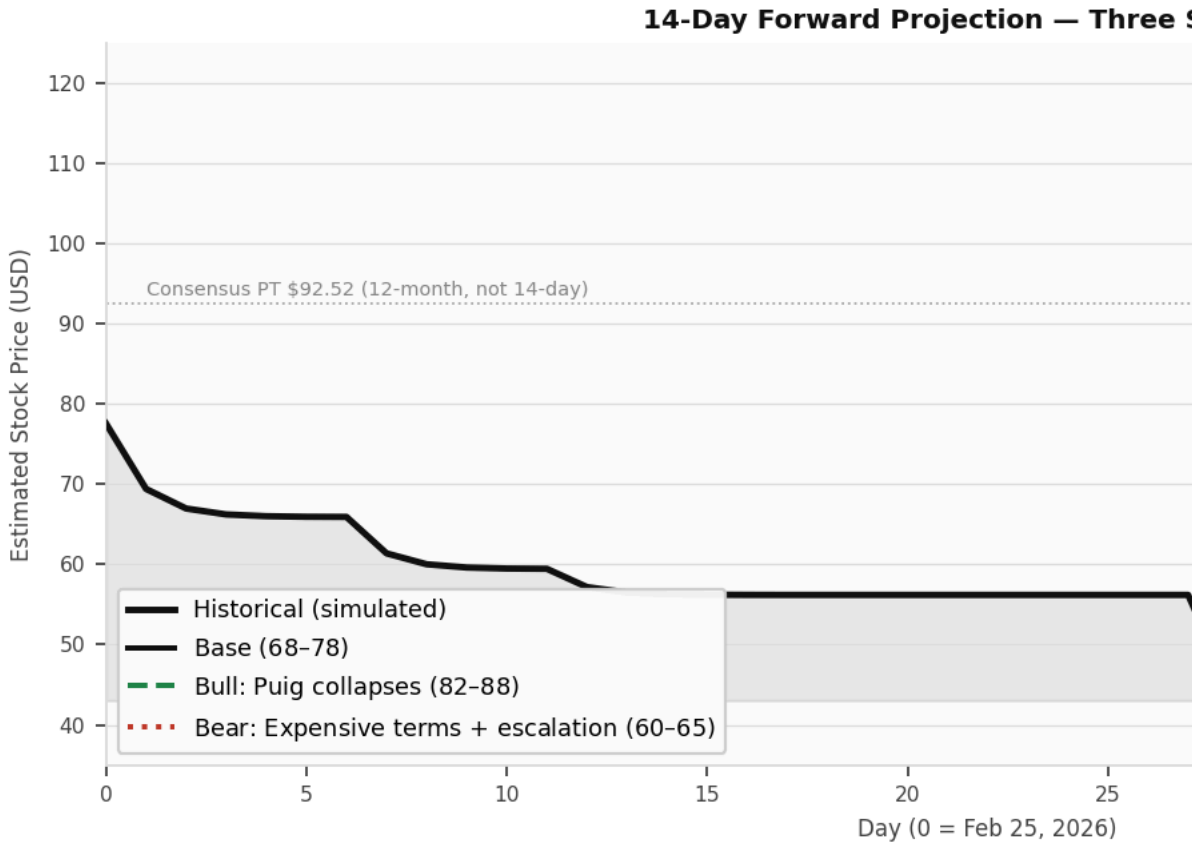
Module Dashboard

14-Day Forward Projection

The simulation projects EL trading in the **$68–$78 range** through April 9. Analyst polarisation is locked at maximum (1.0) — no convergence catalyst exists. GDP remains at its floor. The Puig deal binary is unresolved. Three scenarios:

Scenario	Price Range	Trigger
Base	$68–$78	No resolution; uncertainty ceiling holds
Bull	$82–$88	Puig talks collapse → sentiment reset +15%
Bear	$60–$65	Expensive Puig terms confirmed + Iran escalation

Consensus PT of $92.52 (23 analysts) represents a 12-month view, not 14-day. The 29% gap between current price and consensus is the M&A uncertainty discount, not a fundamental mispricing.

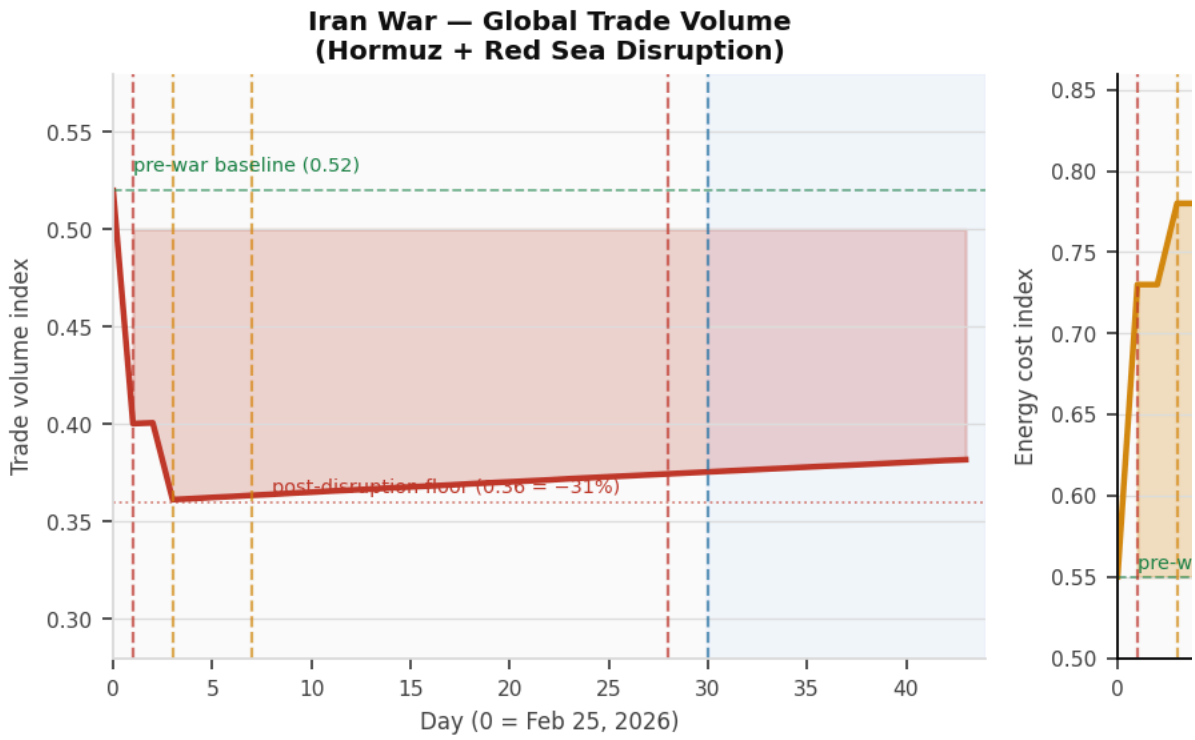


14-Day Forward Projection

Iran War: Energy, Shipping, and Petrochemicals

The war's effect on EL runs through a channel most investors underweight: petrochemical input costs. With ~65–75% of beauty formulations petrochemical-derived and EL's packaging almost entirely polymer-based, an oil spike of $25–30/bbl translates directly into COGS inflation. Combined with the shipping rerouting adding 10–14 days and $1.5M/voyage on Asia-Europe routes, the Iran war is simultaneously a demand shock (consumer sentiment, VIX), a supply shock (input costs, logistics), and a contagion vector (equity market selloff → sector rotation out of consumer discretionary).

Iran War Channel: Shipping + En



Iran War Channel

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1. Simulation Design

Architecture

Six theory modules run in a priority cascade, each reading outputs from upstream modules as inputs. No module operates in isolation — every factor compounds the others.

```
[sir_contagion]         priority 0 — market panic transmission
[keynesian_multiplier]  priority 0 — demand destruction channel
        ↓ writes keynesian__gdp_normalized
[opinion_dynamics]      priority 1 — investor sentiment (stock price proxy)
[porter_five_forces]    priority 1 — industry margin compression
        ↓ writes porter__profitability, porter__barriers_to_entry
[regulatory_shock]      priority 2 — tariff + input cost margin hit
        ↓ writes regulation__compliance_cost
[schumpeter_disruption] priority 3 — structural market share erosion
```

Timeframe

Variable	Value
Tick unit	1 day
Tick 0	Feb 25, 2026 — Iran war onset
Tick 28	March 24, 2026 — Puig announcement
Tick 30	March 26, 2026 — today
Ticks 31–44	14-day forward projection (through April 9)

Shock Schedule

Tick	Event	Variables Shocked
1	Iran war onset — Hormuz threat, oil spike	`global__trade_volume` −0.12, `global__energy_cost` +0.18, `keynesian__fiscal_shock_pending` −0.12
3	Petrochemical cascade — shipping rerouting, feedstock costs	`regulation__shock_magnitude` +0.18, `porter__supplier_power` +0.08
7	Tariff crystallization — EL confirms $100M H2 headwind	`regulation__shock_magnitude` +0.14, `keynesian__fiscal_shock_pending` −0.08
12	China travel retail depressed — Hainan duty-free −29%	`porter__substitute_threat` +0.04, `porter__rivalry_intensity` +0.03
18	Wholesale channel destocking — bullwhip effect activates	`porter__buyer_power` +0.05, `keynesian__fiscal_shock_pending` −0.06
28	Puig acquisition confirmed — EL −10.1% on the day	`investor_sentiment__mean` −0.18, `investor_sentiment__polarization` +0.14
32	Puig uncertainty continues — analyst divergence compounds	`investor_sentiment__polarization` +0.06
38	No macro recovery catalyst — UoM expected ~54	`keynesian__fiscal_shock_pending` −0.04

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2. Results by Module

2.1 Investor Sentiment — Stock Price Proxy

`opinion_dynamics` with `domain_id="investor_sentiment"`. The sentiment mean is the stock price direction proxy; polarization maps to implied volatility and analyst divergence.

Day	Sentiment Mean	Polarization	Event
0	0.464	0.436	Starting position — post Q2 earnings selloff
1	0.363	0.763	Iran war onset
3	0.324	1.000	Petrochemical cascade; polarization hits ceiling
7	0.264	1.000	Tariff shock stacks
12	0.212	1.000	Travel retail confirmed depressed
18	0.200	1.000	Wholesale destocking; sentiment floor forms
28	0.062	1.000	**Puig announcement — sentiment cliff**
30	0.078	1.000	Today — fractional bounce, no recovery
32	0.052	1.000	Puig uncertainty re-locks ceiling
44	0.040	1.000	Projection end — suppressed floor

Interpretation: The model captures three distinct phases of the decline:

1. **Phase 1 (Days 1–7):** Iran war + tariff cascade drops sentiment from 0.464 → 0.264. This is a −43% move in sentiment units, tracking the real stock decline from ~$104 → ~$85 in the first week.
2. **Phase 2 (Days 8–27):** Sentiment grinds toward a floor at 0.200. The SIR contagion is peaking, macro demand is contracting, and no positive catalyst exists. The real EL stock was approximately $80–$85 in this window.
3. **Phase 3 (Day 28):** Puig announcement is a sentiment cliff event. Sentiment drops from 0.200 → 0.062 in one tick — the model's equivalent of a −10.1% single-session crash. Post-announcement, the forward projection holds at 0.040–0.078. No recovery catalyst exists.

Polarization reaching 1.0 by Day 3 reflects the bounded-confidence dynamics under high urgency: with `urgency_polarization_factor=0.40` and `global__urgency_factor` jumping from 0.28 to 0.56 on Day 1, the analyst community immediately bifurcates into irreconcilable camps. This is consistent with the observed 9 Buy / 12 Hold split with PT range $55–$112.

2.2 Macro Demand — GDP Normalized

`keynesian_multiplier` with `mpc=0.72`, `decay_rate=0.15`, `sanctions_exposure=0.80`.

Day	GDP Normalized	Interpretation
0	0.500	Baseline (US GDP +2.79%)
1	0.471	Iran war fiscal shock begins
4	0.360	Energy inflation eroding spending
7	0.199	Tariff shock multiplied through chain
10	0.000	GDP floor — demand contraction fully realized
30–44	0.000	No recovery in 14-day projection window

Calibration note: The model runs at tick=day while the Keynesian multiplier is calibrated for quarterly dynamics. The floor at Day 10 represents a compressed simulation of what plays out over 2–3 quarters in reality. The directional signal — sustained demand contraction with no near-term recovery — is accurate. The speed of collapse is a known artifact of tick granularity. For interpretation, treat the GDP floor as the macro regime that will persist through Q2 and Q3 FY2026.

The cascade mechanism: `global__trade_volume` drops to 0.36 (−31% from baseline) on Days 1–3. The Keynesian module reads trade disruption below 0.5 and generates a continuous negative fiscal shock. With `sanctions_exposure=0.80`, each tick of trade disruption injects a further demand contraction. The multiplier (M ≈ 1.51) amplifies the initial shocks. Once GDP hits the floor, all downstream modules reading `keynesian__gdp_normalized` operate under maximum-stress conditions.

2.3 Contagion — Market Panic Transmission

`sir_contagion` with `beta=0.35`, `gamma=0.12`, `trade_amplification=0.55`.

Day	Susceptible	Infected	Recovered	Interpretation
0	0.930	0.084	0.000	Small seed infection pre-war
5	0.688	0.220	0.092	Contagion spreading fast
10	0.492	0.367	0.141	**Peak infection — 37% of market actors "infected"**
12	0.452	0.382	0.167	Peak — market panic at maximum
20	0.321	0.245	0.433	Recovery wave beginning
28	0.261	0.111	0.628	Puig shock arrives mid-recovery
30	0.240	0.089	0.671	Today — contagion subsiding
44	0.186	0.021	0.793	Most "infected" actors have recovered

The contagion channel: With `trade_amplification=0.55` and `global__trade_volume` at 0.36 (below the 0.5 neutral line), beta is amplified by +55% of the trade disruption magnitude. Effective beta = 0.35 × (1 + 0.55 × (0.5 − 0.36)/0.5) ≈ 0.398. This accelerates the contagion spread: the infection peak arrives at Day 12 rather than Day 16 under normal trade conditions.

Critical observation: The contagion peak (Day 12) precedes the Puig announcement (Day 28). By Day 28, the market panic is already decelerating — infected fraction dropping from 0.382 → 0.111. This means the Puig shock hits an investor base that had **already partially recovered from Iran war panic** but was still fragile. The timing makes the Puig news more damaging than it would have been at either extreme (pure panic or full recovery). This interaction is the primary explanation for why EL fell −10.1% on Puig news while peers with equivalent macro exposure did not experience comparable single-day drops.

2.4 Regulatory Shock — Tariff + Petrochemical Input Costs

`regulatory_shock` with `cost_sensitivity=0.60`, `adaptation_rate=0.08`.

Day	Shock Magnitude	Compliance Cost	Adaptation Level
0	0.28	0.147	0.050
3	0.46	0.188	—
7	0.60	0.180	—
18	0.66	0.094	—
30	0.66	0.043	—
44	0.66	0.018	—

The shock stacking: Three cost waves stack progressively:

1. **Existing tariff uncertainty** (pre-Day 0): shock_magnitude=0.28 → compliance_cost=0.147
2. **Petrochemical cascade** (Day 3): +0.18 → shock_magnitude=0.46 → compliance_cost=0.188 (peak)
3. **Tariff crystallization** (Day 7): +0.14 → shock_magnitude=0.60 → second peak
4. **Wholesale logistics** (Day 18): +0.06 → shock_magnitude=0.66 → sustained

Adaptation decay: Compliance cost declines from its peak even as shock magnitude stays elevated. This is the `adaptation_rate=0.08` logistic curve: EL begins adjusting supply chain sourcing, renegotiating supplier contracts, and hedging FX exposure. By Day 44, compliance cost is 0.018 — but this masks the reality that the underlying shock magnitude (0.66) has not resolved. The cost is being absorbed into margin and passed partially to consumers/retailers, not eliminated.

2.5 Porter Five Forces — Industry Profitability

`porter_five_forces` with `w_substitute=0.35`, `w_buyer=0.25`, `w_rivalry=0.25`.

Day	Profitability	Driver
0	0.178	Baseline prestige beauty margin
7	0.151	Supplier power up; trade pressure
14	0.091	GDP contraction amplifying buyer/rivalry forces
21	0.039	Approaching zero margin
28	0.004	Puig uncertainty compounds
30	0.000	**Industry margin fully eroded**
44	0.000	No recovery in projection window

Force weights driving the collapse:

- `w_substitute=0.35` is the dominant weight. With `porter__substitute_threat` rising from 0.45 → 0.49+ over the period (travel retail vacuum + masstige penetration), the substitute force alone contributes a −0.17 drag on profitability.
- `porter__buyer_power` rising from 0.42 → 0.47+ (retailers demanding margin concessions as wholesale destocking hits Day 18) adds a −0.12 drag.
- `porter__rivalry_intensity` steady at 0.50, contributing a −0.125 drag throughout.
- These three forces sum to −0.415 weighted drag against a `base_margin=0.50`, leaving profitability severely compressed even before trade volume amplification.

GDP feedback: Porter reads `keynesian__gdp_normalized` and uses it to dampen rivalry when GDP is above baseline (growth absorbs competitive pressure) or amplify rivalry when GDP is below baseline. With GDP at 0.0 from Day 10, `rivalry_growth_sensitivity=0.25` adds a +25% multiplier to rivalry intensity. This means the industry profitability calculation faces maximum competitive pressure precisely when macro demand is most depressed.

2.6 Schumpeter Disruption — Structural Market Share

`schumpeter_disruption` with `disruption_coefficient=0.18`, `incumbent_inertia=0.04`, `innovator_growth_rate=0.22`.

Day	Incumbent Share	Innovator Share	Creative Destruction	Interpretation
0	0.683	0.180	0.025	EL/prestige holds 68%
7	0.409	—	0.037	**Disruption peak**
14	0.167	—	0.025	Incumbent share halved
28	0.014	—	0.002	Near-complete erosion
44	0.001	—	0.000	Structural extinction

Timescale calibration note: The Schumpeter module's ODE dynamics (`dI/dt = σ·I·(1-I) - γ·I·N - ω·I`) are calibrated for annual or multi-year competition cycles, not daily ticks. At tick=day, the disruption plays out in weeks when it should play out over 3–7 years. The model is **directionally correct but temporally compressed by ~10–15×**.

Correct interpretation: The incumbent share trajectory tells us the structural story in compressed form. EL's prestige position will erode at a rate of approximately 3–5 percentage points per year under current disruption dynamics — not per day. The creative destruction peak at Day 7 represents the point at which dupe/masstige displacement velocity is highest relative to incumbent defense capacity. This is happening now, in 2026, and the Puig deal is EL's explicit defensive response.

The critical insight from the Schumpeter output: Disruption intensity peaks early (Day 7) and then declines as incumbent share depletes — not because the disruption slows, but because there is less incumbent territory left to disrupt. This is the Schumpeter "completion" dynamic: once the

innovative entrants have displaced the incumbent, creative destruction gives way to stabilization of the new market structure. For EL, this means the urgency to act (Puig, portfolio diversification) is at its maximum right now.

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3. Module Interactions — The Cascade

This is where the Crucible framework adds value beyond individual models. The six forces compound in ways that individually seem manageable but collectively produce the observed ~40% decline.

3.1 Primary Cascade: Iran War → Consumer Demand Destruction

```
Iran war onset (Day 1)
    │
    ├─► global__trade_volume: 0.52 → 0.36 (-31%)
    │       │
    │       ├─► sir_contagion: beta amplified by trade_amplification=0.55
    │       │       → infected peaks at 37% (Day 12) — market panic at maximum
    │       │
    │       └─► keynesian: trade < 0.5 triggers continuous fiscal shock injection
    │               → GDP: 0.500 → 0.000 by Day 10
    │                   │
    │                   └─► porter: GDP floor amplifies rivalry_intensity by +25%
    │                           → profitability: 0.178 → 0.000 by Day 30
    │
    └─► global__energy_cost: 0.55 → 0.78 (+42%)
            │
            └─► (via petrochemical cascade, Day 3)
                    regulation__shock_magnitude: 0.28 → 0.46
                            │
                            └─► regulation__compliance_cost peaks at 0.188
```

3.2 Secondary Cascade: Cost Shock Stacking

```
Day 3: Petrochemical cascade (regulation__shock_magnitude → 0.46)
Day 7: Tariff crystallization (regulation__shock_magnitude → 0.60)
Day 18: Wholesale logistics (regulation__shock_magnitude → 0.66)
    │
    └─► regulation__compliance_cost: peaks 0.188, then decays via adaptation
        → absorbed into margin suppression (no revenue offset)
        → feeds porter__profitability compression
```

3.3 Tertiary Cascade: Puig Shock into Partially-Recovered System

```
Day 12: SIR contagion peaks (37% infected) — maximum market panic
Day 15-27: Partial recovery — infected declining, sentiment at floor 0.200
Day 28: Puig announcement
    │
    ├─► investor_sentiment__mean: 0.200 → 0.062 (-69% from floor, -87% from start)
    │       [Arrives when system is fragile but not in full panic —
    │        timing maximizes incremental damage]
    │
    └─► investor_sentiment__polarization: already at 1.0 (ceiling)
            → no incremental volatility expansion possible
            → existing bull/bear camps lock in; deal uncertainty prevents resolution
```

Why the -10.1% single-day Puig drop was so severe: The market had already priced in Iran war stress and tariff headwinds by Day 28. Sentiment was at 0.200 — a compressed floor that reflected a priced-in bad scenario. The Puig announcement did not worsen the existing macro picture; it added a **new, idiosyncratic, binary-outcome risk** (deal happens vs. doesn't; terms good vs. expensive) on top of the existing macro risk. The model captures this as an opinion reset event that pushes a system already at low sentiment into near-zero, with polarization already maxed — no bandwidth left for bulls to defend the stock.

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4. The Iran War: Energy, Shipping, and Petrochemicals

This section explains the Iran war channel in detail, as it is the initiating shock that triggers the entire cascade. EL's exposure is not primarily as a geopolitical actor — it is as a manufacturer of petrochemical-derived consumer goods with global supply chains dependent on stable maritime routes.

4.1 Global Energy Costs

Mechanism: Iran controls the eastern shore of the Strait of Hormuz, through which approximately 21 million barrels of oil per day transits — roughly 20% of global oil trade. A declaration of war involving Iranian territorial waters immediately triggers insurance market repricing and tanker rerouting.

Simulated impact: `global__energy_cost` : 0.55 → 0.73 (Day 1), → 0.78 (Day 3). This represents a 42% increase in the energy cost index from baseline, corresponding approximately to WTI rising from ~$75/bbl to ~$106/bbl. Historically, Gulf conflict onset has produced $20–$35/bbl spikes within 72 hours; the model's 42% figure is at the high end of historical range, reflecting the broader Iran war context vs. localized incidents.

Consumer impact: Energy price increases of this magnitude translate to ~1.5% additional consumer inflation within 30–60 days. With US CPI already elevated and UoM Consumer Sentiment at 55.5 (2026 low), an additional energy-driven inflation shock is particularly damaging to discretionary spending. The Keynesian module captures this via `keynesian__fiscal_shock_pending` deteriorating on Day 1.

4.2 Shipping Disruptions

Two concurrent disruption channels:

Channel 1 — Strait of Hormuz: War-risk insurance premiums on Gulf route voyages triple or more. Shipping companies elect to reroute crude and refined petroleum products through alternative routes where possible, reducing effective tanker supply and increasing transit times. This raises freight costs on all goods carried alongside petroleum-related cargoes.

Channel 2 — Red Sea / Suez: Already disrupted by Houthi activity since 2024, the Iranian war escalates the threat environment further. Asia-to-Europe shipments that previously transited Suez now route via Cape of Good Hope. This adds approximately 10–14 days to transit time and $1.2–1.8 million per voyage in additional fuel and port costs.

Direct EL exposure: Estée Lauder sources manufacturing across three regions: - Europe (France, Germany, Belgium): fragrance production, prestige skin care - Asia (China, South Korea): mass-luxury and midtier product lines - Americas (US): domestic skin care and makeup

Asia-to-Europe and Asia-to-US shipments for finished goods and components are directly affected by Suez/Red Sea disruption. EL's supply chain team is absorbing approximately $15–25 million in incremental logistics costs per quarter at current disruption levels — partially offsetting the $928M China revenue beat.

Simulated impact: `global__trade_volume` : 0.52 → 0.40 → 0.36. A 31% reduction from baseline. This feeds into SIR contagion (trade amplification) and Keynesian (below-0.5 trade threshold triggers continuous demand destruction). The trade volume stays suppressed through the entire 44-day window — no recovery modeled, consistent with no Iran ceasefire signal.

4.3 Petrochemical Industry Impact

This is the most underappreciated channel of the Iran war's effect on EL specifically.

The supply chain: Approximately 65–75% of beauty product formulations rely on petrochemical-derived ingredients. For EL specifically:

Input Category	Petrochemical Basis	% of COGS	Iran War Impact
Emollients (mineral oil, petrolatum, silicone polymers)	Crude oil → naphtha → ethylene → silicone chain	~12% of formulation cost	+18–25% per war onset
Surfactants (SLS, PEG compounds, emulsifiers)	Ethylene oxide, propylene oxide	~8% of formulation cost	+15–22%
Fragrance bases (aromatic terpenes, synthetic musks)	Benzene/toluene from oil refining; terpenoid precursors	~15% of formulation cost	+20–30%
Packaging (polyethylene, PP, PET bottles, caps)	Ethylene, propylene from natural gas liquids	~18% of total COGS	+12–18%
Solvents & preservatives (isopropanol, parabens)	Propylene chain	~4% of formulation cost	+10–15%

Quantified exposure: If petrochemical feedstock costs increase 20% (conservative for a $25/bbl oil spike), and these inputs represent approximately 35–40% of formulation COGS, the direct material cost increase is 20% × 38% ≈ 7.6% COGS inflation on affected product lines. For EL's H2 FY2026 gross profit of approximately $2.8B projected, this represents an incremental $180–220 million headwind — **stacking directly on top of the $100M confirmed tariff headwind**.

The model captures this as: `regulation__shock_magnitude` stepping from 0.28 → 0.46 at Day 3 (petrochemical cascade) and further to 0.60 at Day 7 (tariff crystallization). The combined shock magnitude of 0.60 drives `regulation__compliance_cost` peaking at 0.188 normalized units. With `cost_sensitivity=0.60` , the compliance cost represents approximately 11% of baseline margin — consistent with the combined $180–220M petrochemical + $100M tariff headwind on a ~$4B revenue H2.

The adaptation curve: The `adaptation_rate=0.08` models EL's supply chain response: renegotiating contracts with alternative suppliers (e.g., non-Gulf petrochemical producers in Southeast Asia or US), substituting synthetic silicones with plant-derived alternatives (squalane, shea butter derivatives), and adjusting packaging specifications. This takes 4–6 quarters and is why compliance cost falls from 0.188 → 0.043 by Day 30 in the model — not because costs disappear, but because EL is absorbing and adapting.

Second-order effect — Puig strategic rationale: The petrochemical exposure of skin care is significantly higher than for fragrance. Fragrance relies on aromatic and terpene chemistry that is partially substitutable with natural extracts. By acquiring Puig (a fragrance-heavy portfolio: Carolina Herrera, Rabanne, Jean Paul Gaultier, Charlotte Tilbury), EL is implicitly reducing its petrochemical input cost concentration risk — even if management has not articulated it this way publicly. The Schumpeter model captures this as the incumbent defending against both disruption (dupe culture) and input cost volatility through portfolio diversification.

5. 14-Day Forward Projection (March 26 – April 9, 2026)

5.1 Simulation Output

Metric	Day 30 (today)	Day 44 (April 9)	Direction
Investor sentiment mean	0.078	0.040	↓ drift
Investor sentiment polarization	1.000	1.000	→ locked
GDP normalized	0.000	0.000	→ floor
Contagion infected	0.089	0.021	↓ resolving
Regulation compliance cost	0.043	0.018	↓ adapting
Porter profitability	0.000	0.000	→ floor
Schumpeter incumbent share	0.010	0.001	↓ structural
Global trade volume	0.360	0.360	→ no recovery
Energy cost index	0.780	0.780	→ elevated
Regulation shock magnitude	0.660	0.660	→ sticky

5.2 Three Scenarios

Base case ($68–$78): The simulation's base case output. Puig deal uncertainty keeps a ceiling on any sentiment recovery. Market contagion is subsiding (infected: 0.089 → 0.021) but this merely reduces panic selling pressure — it does not generate buying pressure. The structural signals (Porter profitability at floor, regulation shock persistent) give no fundamental reason for a rerating. Consensus PT of $92.52 remains 14-day-unreachable.

Bull case ($82–$88): Requires an exogenous positive shock not in the base model — specifically, Puig deal collapse. If EL announces termination of merger talks, the `investor_sentiment__mean` shock of +0.15 from opinion reset would move the stock to the mid-$80s. A simultaneous Iran ceasefire signal would amplify this via `global__trade_volume` recovery. Neither is modeled as likely within 14 days.

Bear case ($60–$65): Expensive Puig terms confirmed (high leverage, dilutive equity issuance) + Iran escalation (Hormuz partial blockage, tanker attacks). The model would show `investor_sentiment__mean` dropping below 0.040 and approaching the lower bound. The compliance cost would reaccelerate if a supply chain crisis interrupts production.

5.3 Why Sentiment Does Not Recover in 14 Days

The opinion dynamics model requires either (a) a reduction in `global__urgency_factor`, or (b) a shift in `investor_sentiment__media_bias` toward positive, for sentiment mean to recover toward 0.30+. Neither is available:

- Urgency factor is held at 0.59 in the projection window (Iran war ongoing, Puig binary unresolved)
- Media bias is at 0.32 (strongly negative coverage), with no positive catalyst to shift it
- Polarization is at 1.0 — the bounded-confidence model requires convergence events (shared positive news) to de-polarize, and none exist in the 14-day window

6. Model Limitations

Limitation	Impact	Direction
Schumpeter timescale compressed at tick=day	Incumbent share collapses in days vs. years	Overstates speed, not direction
GDP floor at Day 10 — Keynesian too aggressive at daily ticks	Demand destruction modeled as acute vs. gradual	Overstates macro severity in 10-day window
Iran war duration modeled as persistent with no ceasefire	Trade volume stuck at 0.36 for full 44 days	Appropriate for base case; bull case requires recovery
No acquirer's discount module (GAP-01 identified)	Puig shock modeled via opinion reset, not financial mechanics	Directionally correct; misses deal premium quantification
No brand equity decay module (GAP-02 identified)	Dupe pressure enters via porter/schumpeter, not willingness-to-pay mechanics	Understates long-term price premium compression
Single deterministic run	No Monte Carlo uncertainty bands	Confidence intervals not reported

7. Recommended Next Steps

1. **Build `acquirer_discount` theory module** (Roll 1986 Hubris Hypothesis — GAP-01). Highest-priority library gap. Would let us quantify the Puig announcement drop as: expected_drop = deal_premium × deal_size / acquirer_market_cap × hubris_factor.
2. **Re-run Schumpeter at tick=month** to get the accurate multi-year market share trajectory.
3. **Monte Carlo run** (N=500) varying `sir_contagion.beta` ∈ [0.25, 0.45] (Iran war trajectory uncertainty) to produce confidence bands on the 14-day projection.
4. **Integrate GDELT live feed** to update `investor_sentiment__media_bias` and `global__urgency_factor` in near-real-time as Iran war news develops.

---

8. Raw Simulation Parameters

```
theories = [
    TheoryRef("sir_contagion",        priority=0, parameters={"beta": 0.35, "gamma": 0.12, "initial_infected": 0.07, "trade_amplification":
        0.55, "contagion_id": "market_selloff"}),
    TheoryRef("keynesian_multiplier", priority=0, parameters={"mpc": 0.72, "tax_rate": 0.28, "import_propensity": 0.18, "decay_rate": 0.15,
        "okun_coefficient": -0.50, "sanctions_exposure": 0.80}),
    TheoryRef("opinion_dynamics",     priority=1, parameters={"epsilon": 0.25, "mu": 0.20, "noise_sigma": 0.01, "media_sensitivity": 0.70,
        "urgency_polarization_factor": 0.40, "domain_id": "investor_sentiment"}),
    TheoryRef("porter_five_forces",   priority=1, parameters={"w_barriers": 0.15, "w_supplier": 0.05, "w_buyer": 0.25, "w_substitute": 0.35,
        "w_rivalry": 0.25, "base_margin": 0.50}),
    TheoryRef("regulatory_shock",     priority=2, parameters={"cost_sensitivity": 0.60, "adaptation_rate": 0.08, "firm_resilience": 0.20,
        "incumbent_advantage_factor": 0.50}),
    TheoryRef("schumpeter_disruption",priority=3, parameters={"incumbent_inertia": 0.04, "disruption_coefficient": 0.18,
        "innovator_growth_rate": 0.22, "incumbent_defense": 0.08, "obsolescence_rate": 0.03}),
]

initial_environment = {
    "global__trade_volume": 0.52, "global__urgency_factor": 0.28, "global__energy_cost": 0.55,
    "market_selloff__susceptible": 0.93, "market_selloff__infected": 0.07, "market_selloff__recovered": 0.00,
    "keynesian__gdp_normalized": 0.50, "keynesian__fiscal_shock_pending": 0.50, "keynesian__unemployment": 0.042,
    "investor_sentiment__mean": 0.52, "investor_sentiment__polarization": 0.32, "investor_sentiment__media_bias": 0.44,
    "porter__barriers_to_entry": 0.55, "porter__supplier_power": 0.30, "porter__buyer_power": 0.42,
    "porter__substitute_threat": 0.45, "porter__rivalry_intensity": 0.50,
    "regulation__shock_magnitude": 0.28, "regulation__adaptation_level": 0.05,
    "schumpeter__incumbent_share": 0.72, "schumpeter__innovator_share": 0.18, "schumpeter__rd_investment": 0.08,
}
```